Biochemistry Cumulative Examination
October 24, 2008

Carbohydrates are an important class of biochemicals. Answer the following questions about them. Keep your answers short and to the point.

1. Arguably the most important carbohydrate is glucose. Draw either L- or D-glucose in the chair form, taking care to show the correct stereochemistry at each position. 15 points.

2. Explain what a pyranose and furanose are. In other words, what is the difference between them? Name one sugar as an example of each. 10 points

3. Explain what a hexose and pentose are. In other words, what is the difference between them? Name one sugar as an example of each. 10 points

4. Sucrose is a disaccharide. It is composed of what two monosaccharides? 10 points

5. Sucrose is often described as being a non-reducing sugar. What is a reducing sugar and why isn't sucrose one? 20 points.

6. Draw a structure for ribose, again taking care to show the correct stereochemistry at each position. 15 points

7. One common modified sugar found attached to proteins is often named using the abbreviation GlcNAc (pronounced glick-nack). What is the full name of this modified sugar? Draw its structure. 10 points.

8. Sugars are often described as being α or β. What does this refer to? Be specific. 10 points